

Lesson 1 · Gradients, Differentiation and Tangents

Learning intention

Differentiate polynomials, and use $f'(x)$ to find the gradient of a curve and the equation of a tangent.

Success looks like

- I can differentiate any polynomial, expanding brackets first.
- I can choose: gradient at a point, point with a gradient, or unknown constant.
- I can write the equation of a tangent in three steps.

Do Now

1 Find the gradient of the line through $(1, 3)$ and $(4, 12)$.

2 Expand and simplify $3x(x - 4)$.

3 Find the equation of the line with gradient -2 that passes through $(3, 5)$.

4 Solve $x^2 - 5x + 6 = 0$.

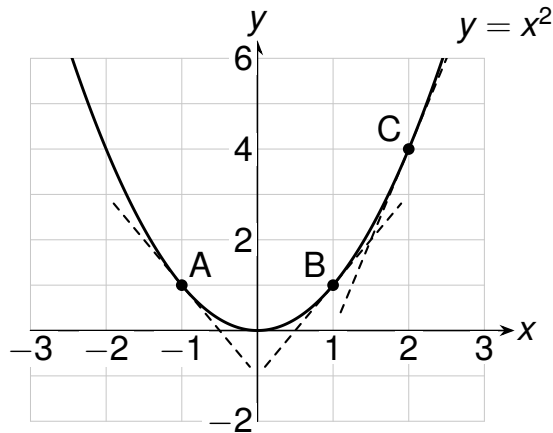
The one idea

A curve's steepness changes from point to point.

The gradient of a curve at a point = the gradient of its **tangent** there.

$f'(x)$ **is a gradient machine**: put in x , get out the gradient.

At A, B and C: going up or down?
Which is steepest?



The power rule

$$\frac{d}{dx}(ax^n) = an x^{n-1}$$

Multiply by the power, then take 1 off the power. Do each term on its own.

term	x^5	$4x^3$	$-6x^2$	$7x$	9
derivative					

Expand brackets first: the rule works on single terms only.

Worked examples · differentiate

1 Differentiate $f(x) = 3x^4 - 5x^2 + 7x - 2$.

Step 1: term by term Step 2: $f'(x) = \dots$

2 Differentiate $y = (2x + 1)(x - 3)$.

Step 1: expand Step 2: $\frac{dy}{dx} = \dots$

Practice

Differentiate.

1 $f(x) = 5x^3 - 4x^2 + x - 9$

2 $y = x(x^2 + 6)$

3 $y = (x - 2)^2$

Three question types

given $\rightarrow$ asked

route

$x \rightarrow$ gradient

differentiate, then **substitute** x **into** $f'(x)$

gradient $\rightarrow$ point

solve $f'(x) = \text{gradient}$ for x ; then y from $f(x)$

gradient at $x \rightarrow$ constant

differentiate with the letter in; substitute x ; solve

The **gradient** comes from $f'(x)$. The y -coordinate comes from $f(x)$.

Worked examples · gradients

- 3 Find the gradient of $f(x) = x^3 - 6x^2 + 4$ at the point where $x = 5$.
- 4 Find the point on $y = x^2 - 7x + 1$ where the gradient is 3.
- 5 The curve $f(x) = ax^2 + 3x - 1$ has gradient 11 at $x = 2$. Find a .

Practice

- 1 Find the gradient of $f(x) = 2x^3 + x^2 - 4x$ at the point where $x = 2$.
- 2 Find the coordinates of the point on $y = x^2 + 4x - 3$ where the gradient is 10.
- 3 The curve $f(x) = kx^3 - 2x$ has gradient 22 when $x = 2$. Find k .

Equation of a tangent

Write it in this order

1. **Point:** y_1 from $f(x)$.
 2. **Gradient:** $m = f'(x_1)$, a number.
 3. **Line:** $y - y_1 = m(x - x_1)$, then $y = mx + c$.
- 6 Find the equation of the tangent to $y = x^3 - 2x + 1$ at $x = 2$.

Worked example · two unknowns

7 The curve $y = ax^2 + bx$ passes through $(1, 5)$, and its gradient at that point is 8. Find a and b .

one equation from the **point** + one from the **gradient**

Parallel lines have equal gradients:

“tangent parallel to $y = 2x + 1$ ” means $f'(x) = 2$.

Practice

- 1 Find the equation of the tangent to $f(x) = 2x^2 - 3x + 4$ at $x = 2$.
- 2 The tangent to $y = x^2 - 4x + 7$ at the point P is parallel to the line $y = 2x + 1$. Find the equation of this tangent.
- 3 The tangent to $y = x^3$ at $x = 1$ meets the curve again. Find the coordinates of this second point.

Ways to lose marks

- L1 Not expanding brackets before differentiating.
- L2 Gradient from $f'(x)$; y -coordinate from $f(x)$.
- L3 $7x \rightarrow 7$ and a constant $\rightarrow 0$.
- L4 The tangent's gradient must be a number, not $f'(x)$.
- L6 "Use calculus": write $f'(x)$ down.

Summary

1. $\frac{d}{dx}(ax^n) = anx^{n-1}$; expand brackets first.
2. Gradient at a point: substitute x into $f'(x)$.
3. Given a gradient: solve $f'(x) = m$; y from $f(x)$.
4. Tangent: point, gradient, then $y - y_1 = m(x - x_1)$.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 to 4: warm-up, about 8 minutes.
- Questions 5 to 10: exam questions, about 22 minutes.
- Show every line of working; write $f'(x)$ each time.

Stuck for more than 3 minutes? Move on and come back.

Homework

Homework sheet: 15 questions, about 60 minutes

- Questions 1 to 12: Achieved practice.
- Questions 13 and 14: Merit.
- Question 15: Extension (Excellence).

Optional extra: StudyTime Guide, “Stop and Check” on pages 10 and 12.